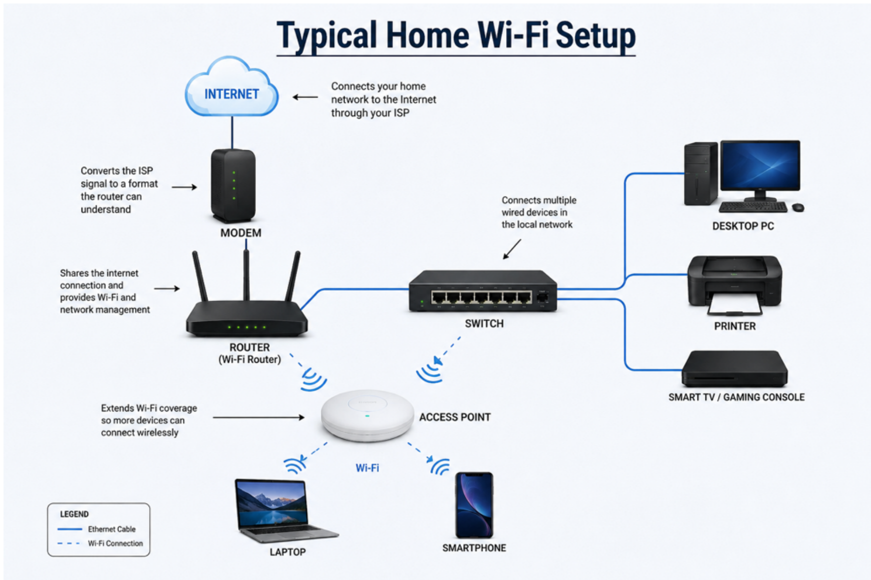
Assignment : Understanding & maintenance of Network.

Day 1 - Assignment

1. **Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.**

⇨



- The **Internet** is provided by the Internet Service Provider (ISP).
- The **Modem** receives the internet signal from the ISP.
- The **Router** distributes the internet to wired and wireless devices.
- The **Switch** connects multiple wired devices such as a desktop PC and printer.
- The **Access Point (AP)** extends the Wi-Fi coverage.
- The **Laptop** and **Smartphone** connect wirelessly through the Access Point.

2. **List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it.**

⇨ Modem main work is to connect home network to the Internet Service Provider for e.g., Getting and new Jio fiber connection to our home.

⇨ Router is used to allow multiple devices to connect wired or wireless and helps to manage network traffic. for e.g., using it at our home to provide wi-fi to devices

⇨ Switch is used connect multiple devices with wired connection in same Local Area Network (LAN). for e.g., used to built computer lab

⇨ Access point is used to extend the wifi range so that more device able to connect in same LAN. For e.g., used to extend in college and offices

Assignment : Understanding & maintenance of Network.

⇨ Network cable is used to transfer data through wired for e.g., connecting a desktop with router/switch

3. **Take photos or find images online (with source links) of at least three types of network cables and write one line describing where each is commonly used.**

⇨ Ethernet cable

Used to connect computers, routers, switches, gaming consoles, and printers in homes and offices.

Coaxial cable

Commonly used for cable television and broadband internet connections.

Fiber Optic cable

Used by Internet Service Providers and data centers for very high-speed internet over long distances

4. **Create a comparison table listing at least four differences between a router and a switch.**

Router	Switch
Connects different networks and provides internet access.	Connects multiple devices within the same Local Area Network (LAN).
Operates at **Layer 3 (Network Layer)**.	Operates at **Layer 2 (Data Link Layer)**.
Uses **IP addresses** to route data between networks.	Uses **MAC addresses** to forward data within the LAN.
Commonly used in homes, offices, and schools to share internet.	Commonly used in offices, computer labs, and server rooms to connect wired devices.
Routes data packets between different networks and the internet.	Forwards data only to the intended device within the same network.

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Assignment : Understanding & maintenance of Network.

Day 2 – Assignment

1. **Draw a labelled diagram of a typical home Wi-Fi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.**

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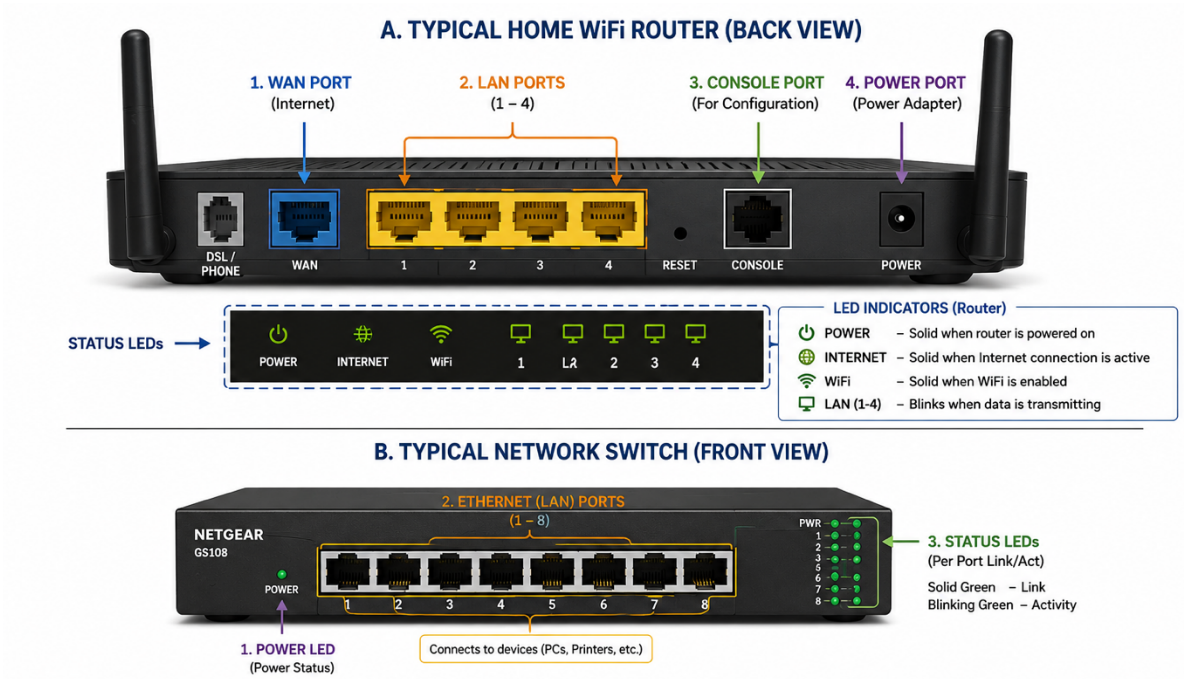


2. **List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.**

⇨

Router	Switch
Connects different networks (such as a home network to the Internet) and routes data between them.	Connects multiple devices within the same Local Area Network (LAN).
Routes data between different networks and controls internet traffic using routing tables.	Forwards data only within the same network, sending it only to the intended destination device.
Uses **IP addresses** to route data between networks.	Uses **MAC addresses** to forward data within the LAN.

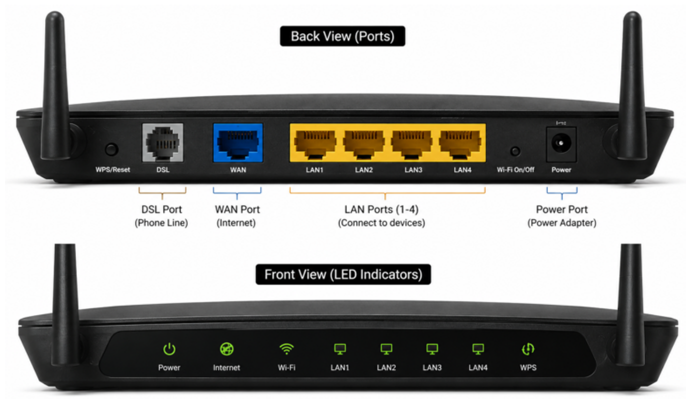
- A **Router** connects different networks, such as your home network to the Internet. It uses **IP addresses** to decide where data should be sent and manages traffic between networks.

Assignment : Understanding & maintenance of Network.

- A **Switch** connects devices within the same Local Area Network (LAN). It uses **MAC addresses** to forward data only to the correct device, making communication faster and reducing unnecessary network traffic.

3. **Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.**

⇨



Power Port	Located on the back, usually at the far right side.	Connects the router to the power adapter to supply electricity.
WAN Port	Located on the back and usually colored blue.	Connects the router to the modem or Internet Service Provider (ISP).
LAN Port 1	Located on the back next to the WAN port.	Connects a wired device such as a desktop computer.
LAN Port 2	Next to LAN Port 1.	Connects another wired device such as a printer or smart TV.
LED Indicators	Located on the front or top panel of the router.	Show the status of Power, Internet, Wi-Fi, and LAN activity. Blinking LEDs indicate data transmission.

4. **Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.**

⇨ While creating café I am going to take 8 Gaming Pc and assemble all the PC with the requirement specs after that I am going to connect ISP with the router through WAN Port so that IP is able to assign in the router and device in the all LAN port is going to connect with the Switch and one switch LAN cable is going to connect with the switch and the switch LAN ports is going to help to established the network in the Gaming café.

Assignment : Understanding & maintenance of Network.

Day 3 – Assignment

1. **Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.**

⇨ **Steps**

   1. Connect your laptop to your home Wi-Fi
   2. Open CMD
   3. Type: ipconfig
   4. Find the default gateway it would be like 192.168.0.1
   5. Open web browser and enter the gateway IP
   6. The router login page will appear

2. **Find Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.**

⇨ **IP :-** 192.168.31.7
⇨ **Subnet:-**255.255.255.0
⇨ **Default gateway:-** 192.168.31.1

3. **Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.**

⇨ Open Network Setting
⇨ Go to wi-fi -> properties -> IPv4
⇨ Select manual
⇨ Enter Ip, subnet, default gateway and preferred DNS
⇨ Save the settings
⇨ Open any website to verify internet connectivity

4. **In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.**

⇨ Login to the router panel Network setting – LAN – DHCP Server
Start range = 192.168.1.100

Assignment : Understanding & maintenance of Network.

- ⇨ End Range = 192.168.1.199
- ⇨ The DHCP server automatically assigns IP addresses to devices connected to the network. If two devices are assigned the same IP address, an IP address conflict occurs. As a result, one or both devices may lose network connectivity, experience internet problems, or be unable to communicate properly with other devices on the network.

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Day 4 – Assignment

1. **List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).**

⇨

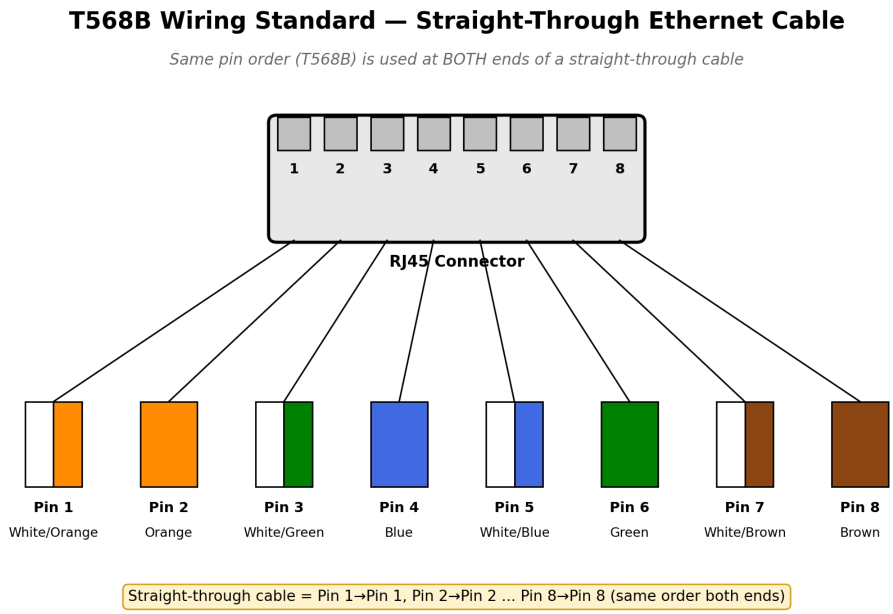
Feature	Ethernet Cable	Fiber Optic Cable
Transmission Medium	Transmits data using **electrical signals** through copper wires.	Transmits data using **light signals** through glass or plastic fibers.
Speed & Distance	Suitable for short distances (up to 100 meters) with speeds up to 10 Gbps (depending on cable category).	Supports much higher speeds and can transmit data over several kilometers without significant signal loss.
Cost	Less expensive and easier to install.	More expensive and requires specialized installation and equipment.

Real-World Examples

- **Ethernet Cable:** Used to connect a **home Wi-Fi router to a desktop PC** or gaming console.
- **Fiber Optic Cable:** Used as the **backbone network in a college campus or data center** to provide high-speed internet.

2. **Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.**

Assignment : Understanding & maintenance of Network.



- In a straight-through cable both ends use the exact same T568B pin order (shown in the diagram above), so pin 1 on one connector goes to pin 1 on the other, pin 2 to pin 2, and so on all the way to pin 8.
- This is different from a crossover cable, where one end is wired T568A and the other T568B. Since I only needed to connect a PC/laptop to a router or switch, a straight-through (T568B → T568B) cable was the right one to make.

3. **Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.**

- **Steps**
  7. Use the stripper blade on the crimping tool to cut and pull off about 1 inch of the outer plastic jacket from the end of the CAT6 cable, being careful not to nick the coloured wires inside.
  8. Untwist the 4 pairs of wires and straighten them out flat.
  9. Arrange the 8 wires side by side in the T568B colour order: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
  10. Use the cutter on the crimping tool to trim all the wires evenly, leaving about half an inch of straight wire.
  11. Push the wires all the way into the RJ45 connector so each coloured wire reaches the front and lines up with its correct pin.
  12. Place the connector into the crimping tool and press the handle down firmly — this pushes the metal pins through the plastic and locks them onto the wires.

13. Gently pull the cable to make sure the connector is firmly crimped and doesn't slide off.
14. Repeat the same process on the other end of the cable, then plug it between a PC and a router/switch to test the connection.

4. **Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Hint: Consider speed, distance, and budget for your answer.**

- For a gaming café with only 10 PCs, I would use Ethernet (CAT6) cables and an RJ45 crimper instead of fiber optics. Since all the PCs are in the same room and the distance between them is short, Ethernet already gives more than enough speed (up to 1 Gbps) and very low latency, which matters more for LAN gaming than extra bandwidth. Fiber optic cables and fiber cleavers are meant for very long distances or data-centre level speeds, and they cost a lot more, so they would be overkill and too expensive for a small 10-PC setup. So Ethernet with a crimper is the more practical and budget-friendly choice here.

Assignment : Understanding & maintenance of Network.

Day 5 – Assignment

1. **Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.**

⇨ **Steps**

2. Connected my laptop and my friend's phone to the same home Wi-Fi network.
3. Checked my friend's phone's IP address from its Wi-Fi settings — it was 192.168.31.15.
4. Opened Command Prompt on my laptop and ran: ping 192.168.31.15
5. Got replies back from the phone within a few milliseconds, which confirmed both devices could communicate on the network.

```
Pinging 192.168.31.15 with 32 bytes of data:

Reply from 192.168.31.15: bytes=32 time=4ms TTL=64

Reply from 192.168.31.15: bytes=32 time=3ms TTL=64

Reply from 192.168.31.15: bytes=32 time=5ms TTL=64

Reply from 192.168.31.15: bytes=32 time=3ms TTL=64


Ping statistics for 192.168.31.15:

    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

⇨ All 4 packets came back with 0% loss, so both devices were properly connected and able to communicate with each other on the same Wi-Fi network.

2. **Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.**

⇨ **Command used:** tracert www.youtube.com (on Windows)

```
Tracing route to www.youtube.com [142.250.194.14]

over a maximum of 30 hops:


  1     2 ms     1 ms     1 ms  192.168.31.1

  2    12 ms    11 ms    10 ms  10.10.10.1

  3    14 ms    13 ms    14 ms  isp-core-router.net
```

```
4    18 ms    17 ms    19 ms  72.14.xxx.xxx

5    20 ms    19 ms    21 ms  108.170.xxx.xxx

6    22 ms    21 ms    23 ms  142.250.xxx.xxx

7    25 ms    24 ms    26 ms  142.250.xxx.xxx

8    28 ms    27 ms    26 ms  142.250.194.14


Trace complete.
```

- ⇨ It took about 8 hops to reach www.youtube.com from my network. The first hop (192.168.31.1) was my own home router, the next couple of hops were inside my ISP's network, and the last few hops were Google's own network before finally reaching the YouTube server.

3. **Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.**

- ⇨ Turned off Wi-Fi on my friend's phone (192.168.31.15) so it disconnected from the network, then ran the same command again: ping 192.168.31.15

```
Pinging 192.168.31.15 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.


Ping statistics for 192.168.31.15:

    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

- ⇨ The error I got was “Request timed out.” This happens because the phone is no longer connected to the Wi-Fi, so it can't reply to the ping request. Sometimes you can instead see “Destination host unreachable”, which means the network couldn't even find a path to that device.

4. **Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.**

- ⇨ Ran arp -a in Command Prompt on my laptop while connected to the home Wi-Fi and noted down the devices below:

IP Address	Device	Belongs To
192.168.31.1	Wi-Fi Router	Network Gateway

192.168.31.7	Laptop	Me
192.168.31.15	Smartphone	Friend
192.168.31.20	Smart TV	Living Room
192.168.31.25	Smartphone	Mom

5. **Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.**

⇨ **Network Connectivity Testing Report**

I utilized my home Wi-Fi network, with a router at 192.168.31.1, to test connectivity between my laptop and a friend's smartphone. My laptop’s IP was determined using ipconfig, revealing it as 192.168.31.7, while the smartphone’s IP was found to be 192.168.31.15. I executed a ping command (ping 192.168.31.15) from my laptop, receiving replies with 3–5 ms round-trip times and 0% packet loss, confirming connectivity between devices. I then conducted a tracert to www.youtube.com, mapping the data path through my router and ISP to Google’s servers, highlighting the difference between ping’s reachability check and traceroute's detailed routing information. A connectivity issue occurred when I turned off my friend's phone, resulting in ping failures, illustrating that a disconnected device cannot respond. This exercise demonstrated the efficacy of ping and traceroute as tools for network communication verification and path analysis.